



MAKING THE INTERNET OF THINGS
**INTEL GALILEO: WHAT WILL
YOU MAKE?**

Integrating the digital world into the physical world is THE NEXT BIG THING. In the near future, we will have billions of interconnected devices monitoring our health, factories, farms, forests, weather conditions, oceans and other natural and man-made resources. While all these monitoring devices will be connected to the Internet, most of them won't directly be used by people but networked to be controlled by computers, with much of the information being exchanged between machines directly and without any human involvement. It is estimated that more than 30 billion devices will be connected to the Internet of Things by 2020 monitoring health, daily activity, environment (to use resources more efficiently) etc.

With the evolution of the maker culture, such systems are being built by everyone and

not just by people with specialized knowledge. The electronics maker community has widely used the Arduino platform for many projects. Makers have created projects such as tree climbing robot, auto plant watering system and even something like email/RSS notifier using the Arduino platform. Complicated projects need more processing power so developers interface multiple boards or interface boards to personal computers to increase processing power. For those unaware of Arduino, it is an Open Source electronics platform with easy-to-use hardware and software aimed at hobbyists, designers and anyone interested in creating interactive things using electronic hardware.

At the core, Arduino has a microcontroller (a small computer on a single chip containing a processor, memory and programmable input/output ports). Programming using the Arduino

software is very simple because of the modified C programming compiler developed by Arduino developers, which can be used to write and upload code to the board. Thus, anyone capable of working out common logic problem can code using Arduino. The Arduino software works on Linux, Windows and Mac OS X platforms.

Arduino has many hardware add-ons called 'shields' that can be used directly with an Arduino board to simplify development. There are add-ons such as:

- GSM Shield (for GSM wireless connectivity)
- Ethernet Shield (for using Ethernet port to connect to internet)
- Wi-Fi Shield (to connect to internet wirelessly)
- Motor Shield (to connect DC, stepper motors etc. via L298 driver)
- Proto Shield (for prototyping of custom circuits)